David Liu

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EDUCATION

Boston University — Boston, MA
Ph.D. in Computer Science, focus on Artificial Intelligence for Audio — Sep 2023 – Present
Advised by Dr. Brian Kulis

University of Wisconsin–Madison — Madison, WI
Bachelor of Science — May 2023
Majors: Computer Science, Data Science, Mathematics — GPA: 3.868/4.000

PUBLICATIONS

Interpretable Destination-Aware Synthesizer Modulation Recovery — Under review, ICASSP 2027
David Liu, Giulio Cengarle, David Cooper, Mark Vinton, Haici Yang — arXiv 2026

Are Seismic Foundation Models More Than Glorified Feature Extractors? — NeurIPS 2026 STODY Workshop
David Liu*, Laura Laurenti*, Brian Kulis, Men-Andrin Meier, Christopher W. Johnson

Does the Foundation-Model Paradigm Transfer to Earthquake Data? — NeurIPS 2026 WiML Workshop
Laura Laurenti*, **David Liu***, Brian Kulis, Christopher W. Johnson

Global PIQA: Evaluating Physical Commonsense Reasoning Across 100+ Languages and Cultures
Tyler A. Chang, …, **David Liu**, … et al. — NeurIPS 2026

** Equal contribution*

PRESENTATIONS

Interpretable Destination-Aware Synthesizer Modulation Recovery — Cambridge, MA
Poster, Speech and Audio in the Northeast (SANE) 2026, MIT — Oct 2026
David Liu, Giulio Cengarle, David Cooper, Mark Vinton, Haici Yang

A State-Space Foundation Model for Raw Seismic Waveforms — New Orleans, LA
AGU Fall Meeting 2025 — Dec 2025
David Liu, Laura Laurenti, Christopher W. Johnson, Brian Kulis

Meter Detection: Computing Audio Features Using Mamba-Based Models — Cambridge, MA
Poster, Speech and Audio in the Northeast (SANE) 2024, Google — Oct 2024
David Liu, Brian Kulis

EXPERIENCE

Dolby Laboratories — Atlanta, GA
Ph.D. Research Intern (Mentor: Haici Yang) — Jun 2026 – Sep 2026

- Built an interpretable inverse-synthesis pipeline that detects which synthesizer parameters are modulated, then recovers each LFO shape and the oscillator waveform, trained through a custom differentiable synthesizer
- Showed that a Gammatone loss and a CQT discriminator improve perceptual quality over the standard multi-scale spectral loss, confirmed in a listening test with 12 expert listeners
- First-author paper under review at ICASSP 2027; poster at SANE 2026

Boston University, Department of Computer Science — Boston, MA
Graduate Research Assistant, AI for Understanding Earthquakes — Aug 2024 – Present

- Developing a state-space encoder–decoder foundation model for raw seismic waveforms
- Built a training dataset from waveforms recorded at monitoring stations near Parkfield, CA
- Work presented at AGU 2025 and two NeurIPS 2026 workshops

WORKSHOPS & REVIEWING

Learning to Listen: Machine Learning for Audio at ICML — Seoul, South Korea

Workshop Co-organizer Jul 2026

AI Heard That! Machine Learning for Audio at ICML Vancouver, BC
Workshop Co-organizer Jul 2025

Reviewer: ISMIR 2026; TMLR (2024 – Present); ICML Workshop on Machine Learning for Audio (2025, 2026)

TEACHING

Boston University, Department of Computer Science Boston, MA
Teaching Fellow, CS 131 Combinatoric Structures (Dr. Peter Golbus) Aug 2023 – May 2024

- Led weekly discussion sections and office hours
- Prepared weekly course outlines, helped plan exams, and organized grading

University of Wisconsin–Madison Math Learning Center Madison, WI
Peer Mentor Oct 2021 – May 2023

- Led group sessions for Calculus, Multivariable Calculus, Linear Algebra, and Analysis students
- Planned review meetings to help students prepare for exams

AWARDS

Zero Robotics NASA/MIT World Competition: World Champion 2019 (First Place)
NFMC Junior Composition Contest: Honorable Mention for National 2019
WSMA Composition Project: First Place for Piano Solo 2016 and First Place for Ensemble 2020
University of Wisconsin–Madison Dean's List

SKILLS

Programming Languages: Python, C, Java, Rust, Haskell, JavaScript, C#, R
Tools: LaTeX, Linux